

Virtual Reality-Based Distraction on Pain and Performance during and after Moderate-Vigorous Intensity Cycling

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ABSTRACT

This experiment measured effects of visual perceptual load (PL) within immersive virtual reality (VR) on exercise-induced pain during cycling. Using a within-subjects design ($n=43$), participants cycled at a perceptually “hard” intensity for 10 minutes without VR (i.e., no PL - NPL) or with VR of low or high PL (i.e., LPL or HPL). Mean quadriceps pain was significantly lower in the NPL condition than either the LPL ($d=0.472$) or HPL conditions ($d=0.391$). Mean cycling performance was significantly greater during the LPL condition. Compared to traditional cycling (NPL), cycling in the LPL condition resulted in greater exercise performance despite greater pain.

Keywords: Virtual reality, cycling, pain intensity, perceptual load, mental effort, anxiety.

Index Terms: Interaction design, emerging technologies, behavioral sciences.

1 INTRODUCTION

Greater attention to pain leads to higher pain intensity (PI) and an absence of habituation over time [1]. Distraction techniques, which inherently draw attention away from pain, decreases pain responses and does result in habituation in a number of situations [1]. Pain reducing effects of distraction on exercise-induced pain are equivocal. Immersive virtual reality (VR) is considered a distraction technique that has previously demonstrated pain-relieving effects during painful, high intensity, interval cycling [2]. There is general agreement that VR’s distracting nature stems from its ability to engage attention (i.e., with high immersion and presence), but those concepts are vague. There is a lack of understanding as to why VR is so distracting which leads to challenges in creating the optimal VR program to relieve pain.

Perceptual load (PL) may be one key component to the highly distractible nature of VR. PL is operationally defined as the number of unique objects present in an environment that may or may not be targets of interest. A greater number of unique items indicates a higher PL [3]. Increased PL attenuates the perception of many types of competing stimuli, including painful stimuli [4], but has not yet been tested on exercise-induced pain specifically.

The primary purpose of this experiment was to better understand the extent to which different levels of PL within a virtual environment relieve exercise-induced quadriceps PI. We hypothesized that PL would have a statistically significant inverse

relationship with PI during cycling, such that a greater PL would relate to lower pain intensity and greater cycling performance.

2 METHODS

This experiment employed a within subjects design wherein all participants were tested under all three PL conditions: none (NPL), low (LPL), and high (HPL). Everyone completed the NPL condition first to familiarize them with a self-paced bout of cycling. The order of the other conditions was randomized using a random number generator in Microsoft® Excel 2010. Based on an a priori power analysis, we planned to recruit 45 healthy participants (aged 18-29). Potential participants were excluded if they reported contraindications to high intensity exercise, $18 > \text{BMI} < 35$, prescription medication use, visual impairments, or a high likelihood of motion sickness. All participants signed an informed consent approved by the Institutional Review Board at the University of Georgia. Compensation was given in either class credit or Amazon gift cards when class credit was unavailable.

All cycling was done on the same electronically braked cycle ergometer (Lode Excalibur). After a five-minute warm-up at a low resistance (20 watts), participants cycled at a self-paced rate for 10-minutes (10SPB) before completing a maximal effort 30-second sprint (30S) [2], followed by a five-minute cool-down (20 watts). Prior to cycling, the researcher explained ratings of perceived exertion (RPE) using Borg’s 6-20 scale [5]. Participants were instructed to work at RPE ~15, or “hard intensity”, during the 10SPB, and to give an “all-out” effort (RPE ~20) for the 30S.

Quadriceps PI was measured at four time points using a well-validated 0-10 category scale with ratio properties displayed in the VR [6]: every three minutes during the 10SPB (E1-E3) and following the 30S (E4). Cycling performance was quantified from the total work (kilojoules) performed during both the 10SPB and 30S. RPE [5], and mental effort [7] were measured with PI.

2.1 Perceptual Load Conditions

As shown in Figure 1, the VR was a straight, dirt, bike path. During the 10SPB only, participants passed by items along the path as they moved forward, including trees, animals, and flowers. Continuously displayed (not memorized), in the central foreground were a subset of items, called “target items”. During the 10SPB only, participants were tasked with identifying those target items by pressing a simple counter. PL was manipulated via the number of unique items, number of target items, and the inter-stimulus interval (time between presentation of items).

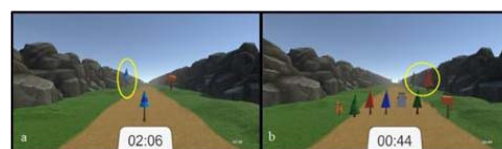


Figure 1: Virtual bike path. The VR participants saw during the 10SPB in the (a) LPL and (b) HPL conditions.

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2.2 Statistical Analysis

A 3x4 condition (NPL, LPL, HPL) x time (E1, E2, E3, E4) repeated measures ANOVA was used to test the effect of condition, time, and their interaction on quadriceps PI during cycling. A series of one-way repeated measures ANOVAs were used to test the effect of each condition on performance in the 10SPB and 30S separately. Huynh-Feldt ϵ adjustments were used when sphericity was violated. Effect size magnitudes were calculated using standardized group differences, $d = (\text{mean } x_1 - \text{mean } x_2) / \text{SD}_{\text{pooled}}$ and partial eta squared values (η^2) for interactions. When the interaction was significant, dependent t-tests were used as post-hoc tests.

3. Results

Forty-three participants completed all testing sessions (20 females). Twenty-two participants completed the LPL condition first. For PI, there were significant main effects of condition ($F=13.458$, $df=1.579, 66.334$, $p<0.001$) and time ($F=113.045$, $df=1.618, 227.683$, $p<0.001$) (Figure 2). At every time point, mean PI was significantly lower in the NPL than in the LPL condition ($t(42)=4.737$, $p<0.001$, $d=0.472$) and HPL condition ($t(42)=3.380$, $p=0.002$, $d=0.391$). However, there was no significant difference in PI between VR conditions ($p=0.314$).

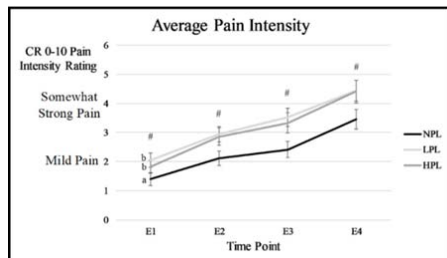


Figure 2: Pain intensity during exercise. Mean ($\pm$ SE) PI data are shown for each of the three PL conditions at four timepoints. (ab) Condition effect ($p<0.05$). #Time effect ($p<0.05$).

There was no difference in performance during the 30S between condition. There was a significant condition effect on work performed during the 10SPB ($F=9.532$, $df=1.585, 66.561$, $p=0.001$) (Figure 3). More work was performed during the LPL than the NPL ($t(42)=2.992$, $p=0.005$) and HPL ($t(42)=5.810$, $p<0.001$) conditions. There was no difference between work performed during the NPL and HPL conditions ($p=0.667$).

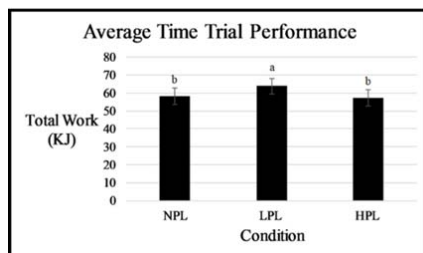


Figure 4: Total work during the time-trial. Mean ($\pm$ SE) total work (kilojoules) data are shown for each of the three PL conditions. (ab) Condition effect ($p<0.05$).

4. Discussion and Conclusion

In contrast to our hypothesis, PI during cycling was greater in the two VR conditions than in the NPL condition. Greater pain during the LPL condition is most likely explained by the greater total work the participants chose to perform during the 10SPB [6]. Greater work, given that the 10-minute time was fixed across condition,

means that the relative intensity was higher in the LPL and this was supported by higher RPE values.

One plausible explanation for higher pain in the HPL condition, based on the data obtained, is the higher mental effort required. One prior experiment that manipulated cognitive load to three levels and measured resting brain blood flow in response to concomitant noxious stimuli found that flow responses were only modulated when cognitive load was low [8]. Other experiments have suggested that the effect of cognitive load on pain varies based on the type of concurrent task [9].

Strengths of this experiment included the specific manipulation and measurement of PL without involving working memory. In addition, with the sample size, within-subjects design and low attrition rate, there was adequate statistical power to detect small effects (0.25 SD). Finally, only one participant reported simulator sickness from the combination of VR and cycling. Improvements to this study include the direct connection between cycling speed and VR movement, and a more accurate measure of success in the item detection task.

Future research on exercise and VR for exercise-induced pain should look to recruit a sample of persons with pre-existing pain and manipulate other factors of VR, such as personal choice, novelty, and competition or reward. In addition, future research should explore the dose-response relationships between exercise characteristics (intensity, duration) and VR characteristics (PL, mental effort) to decrease exercise-induced pain.

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